

◆ VR-Based Product Evaluation System ◆

Instructions (EN)

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◆ INTRODUCTION TO THE PROJECT

This Unity project, titled VR-Based Product Evaluation System, is a tool designed for researchers who wish to conduct product evaluations in Virtual Reality (VR) environments without the need for specialized knowledge in virtual environment development or data management.

The platform allows the collection and analysis of different users' perceptions of a virtual prototype of a specific object through a Semantic Differential, while recording Eye-Tracking (ET) and user position data within an immersive virtual environment. Additionally, it includes a complementary application (.EXE executable) to convert the obtained data into formats that are understandable and easily interpretable by the researcher.

To use this tool optimally, it is necessary to have the Meta Quest Pro Head Mounted Display (HMD), an advanced VR device equipped with ET technology.

This manual is organized into five sections:

- ✓ Getting Started: Explanation of the essential software needed to use the tool, along with detailed instructions on downloading and installing the project.
- ✓ Resources: Description of the project's structure and the location of its various resources to facilitate their identification and use.
- ✓ Setup: Although the project includes pre-designed resources, some must be adapted to specific study cases. This section guides the researcher in modifying the configurable elements.
- ✓ Operation: A detailed explanation of how to use the system after setup, including instructions for its correct operation.
- ✓ Data Analysis: The data obtained in Comma Separated Value (CSV) format must be processed using the application included in the project. This section explains how to transform the information into results useful for the research, such as (1) a map of the user's trajectory, (2) a heatmap of the user, and (3) the total time spent observing each Area of Interest (Aoi) of the product.

◆ FIRST STEPS

1. Installing Unity Hub and Unity

Download and install [Unity](#) (the minimum recommended version for an initial test is 2022.3.46f1) in order to open the project. It is important to make sure that the Android Build Support option is selected during the installation. Additionally, although [Unity Hub](#) is optional, its installation is recommended.

If you have never used Unity, you can find additional information and introductory tutorials [here](#). It is also advised to familiarize yourself with the overall structure of its [interface](#), as some sections will be referenced when locating options, performing specific actions, etc.

Additionally, [Git](#) must also be installed. Once all the software is installed, it is recommended to restart your computer to ensure the proper functioning of the project.

2. Downloading project and importing to Unity Hub

Download the project from [here](#) in ZIP format (Figure 1.a), save it to your PC, and extract the files (Figure 1.b).

Next, open Unity Hub and create an account. Then, click on Add Project from disk in the Projects tab (Figure 2.a). Find the downloaded project and click on the download folder until you see subfolders such as Assets, Logs, and Obj. At that point, click on Add Project or Open (the terminology may vary depending on your computer).

The project will appear in the Unity Hub interface, and you will be able to open it. Unity may prompt you to restart the program after loading the resources, so click Restart as shown in Figure 2.b.

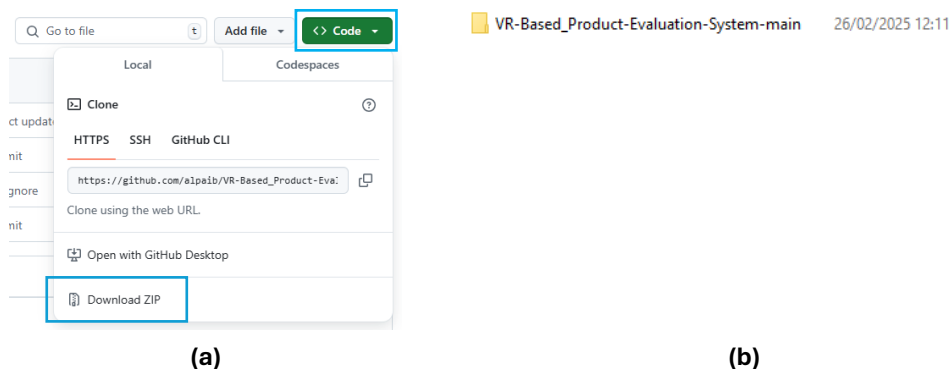


Figure 1. Download the project in ZIP format from GitHub.

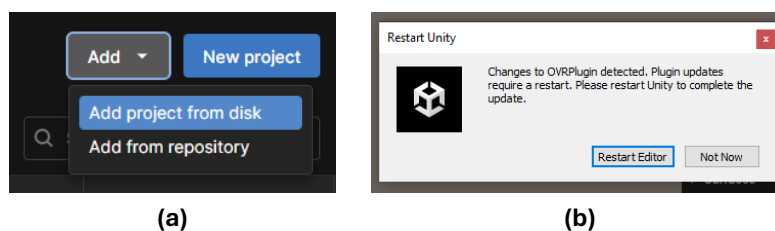


Figure 2. In (a), it shows how to add the project to Unity Hub, and in (b), the pop-up window asking to restart Unity after loading the resources.

◆ RESOURCES

This project includes a series of resources organized into folders to ensure its proper functionality. In this context, Figure 3 shows all these folders. The purpose of this section is to describe each one, which will allow you to easily locate them once the content of each is explained.

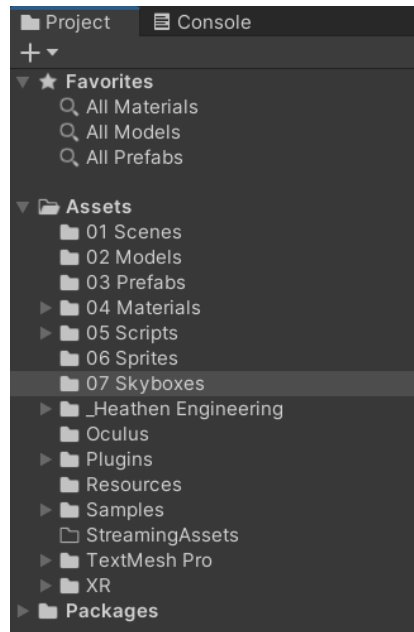


Figure 3. Main folders that make up the Unity project.

1. Scenes

The scenes of the project are located in the "01 Scenes" folder and are as follows:

- **Intro-scene (S-1):** This is the first scene where the user is placed at the beginning of the experience. It takes place in a room within a neutral, low-light environment. During the virtual experience, the user selects their subject code (assigned by the researcher beforehand) on a panel with the help of the Meta Quest Pro controllers.
- **Evaluation-scene (S-2):** This scene depicts a neutral environment with higher lighting. In this scene, the subject can observe a specific product while Eye-Tracking (ET) and position data are collected. Using the Meta Quest Pro controllers, the subject can evaluate the different attributes of the product through a Semantic Differential.

2. Models

The "02 Models" folder contains the 3D models that can later be added to scene S-2. It is recommended that the format be FBX, although other [formats](#) can also be used.

3. Prefabs

A prefab is a file that acts as a template for an object or a set of objects. It allows you to save configurations, components, and object hierarchies so they can be easily reused in different scenes or projects without having to recreate them manually each time. In the "03 Prefabs" folder, there are two important ones:

- **Snapshot:** This element is added to scene S-2 to take an overhead capture of the scene. It is then used to paint the subject's trajectory and generate the heatmap of their path.
- **Vol-Prefab:** This represents a Volume of Interest (Vol), which is a specific region within the 3D space that defines, through its reconfiguration, parts of the product being observed. It will be detected by the gaze in the Meta Quest Pro to collect Eye-Tracking (ET) data..

4. StreamingAssets

The folder named "StreamingAssets" contains the capture taken with the Snapshot prefab. Additionally, it stores the CSV files containing the evaluation, ET, and position data in case the virtual experience is carried out with the HMD connected to the computer once the application is stopped. Each file is saved with a unique naming convention for each subject:

- Code_Eye-tracking (ET data).
- Code_Position (position data).
- Code_Semantic-Differential (evaluation data).

5. Others

5.1. Materials

To ensure good organization of the project, it is recommended that any new materials created for the product being evaluated be stored in the "04 Materials" folder.

5.2. Scripts

The scripts used to develop this project are located in the "05 Scripts" folder. Additionally, within "Data analysis – dist", you will find the application "Interface.exe", which will be used to process the data collected during the virtual experience.

5.3. Sprites

A sprite is a 2D image used in video games and applications to represent objects, characters, backgrounds, or other visual elements. To ensure good organization of the project, it is recommended that all new sprites that are created and/or imported be stored in the "06 Sprites" folder.

5.4. Skyboxes

Similarly, it is recommended that all new skyboxes that are created be stored in the "07 Skyboxes" folder. Additionally, within this folder, you will find the "Dark-Skybox" and "Soft-Skybox" materials, used in S-1 and S-2 respectively, which can be modified without affecting the functionality of the application.

◆ CONFIGURATION

1. Adding user codes

Currently, in S-1, there are alphanumeric codes available for 100 people, generated by combining a letter (A-J) and a number (1-10). If the researcher needs more combinations, they can add more letters or numbers using the "L-Dropdown" (letters) and "N-Dropdown" (numbers) components, which are located within the User Code/Canvas component in the hierarchy (Figure 4.a) in the S-1 scene (intro-scene).

By selecting one of these components, you can add more options in the Dropdown – TextMeshPro field in the inspector (located on the right side of the Unity interface). At the bottom of this component, you will see the "+" (to add) and "-" (to remove) symbols, which will allow you to modify the available options (Figure 4.b).

After making these changes, it is recommended to save the scene and the project (File – Save or Ctrl + S; File – Save Project or Ctrl + Shift + S).

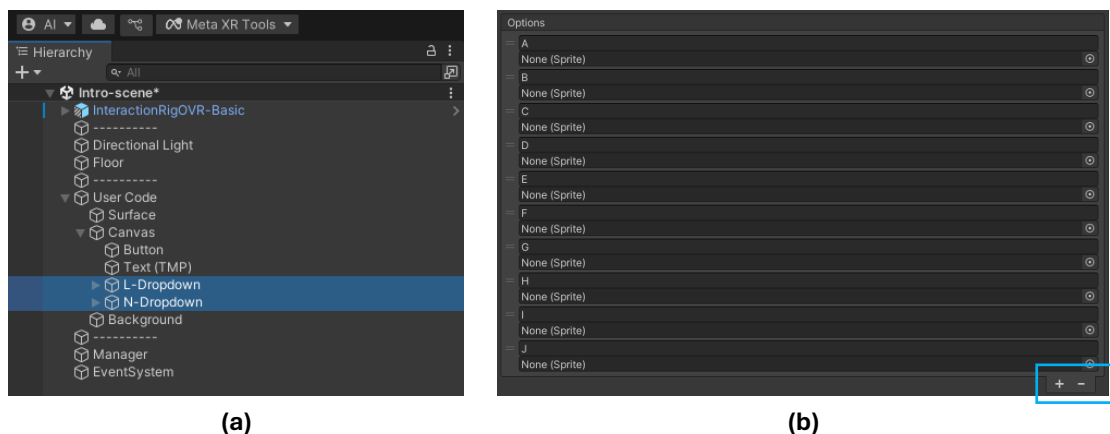


Figure 4. In (a), the components within the Unity hierarchy are shown, and in (b), it shows how to add more items.

2. Introducing a new product

2.1. Import FBX

First, we need a 3D model in FBX format. In the "Project" window (at the bottom of the Unity interface), navigate to the "02 Models" folder. By right-clicking on it, we can select the option Import New Asset... (Figure 5). Then, search for our model on the computer and import it.

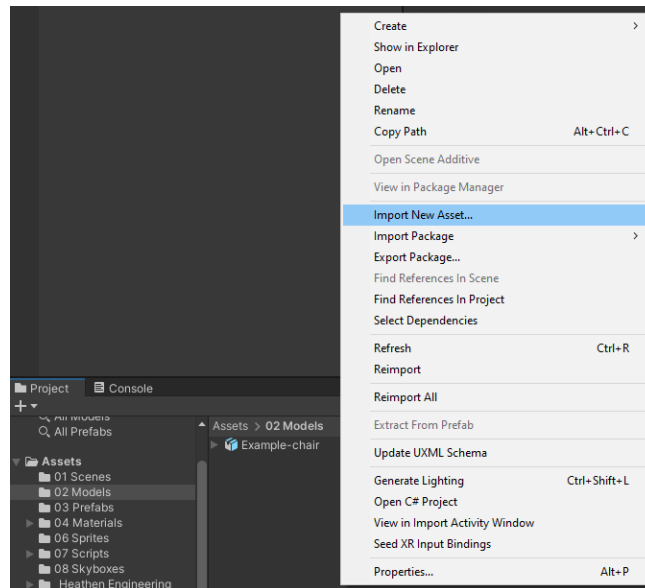


Figure 5. Import FBX to the project.

2.2. Adding model to hierarchy

Secondly, it is necessary to place our model into the "Evaluation-scene" (S-2). To do this, click on the recently imported FBX model, and while holding the click, drag it into the hierarchy onto "ADD PRODUCT HERE". This will make your model a child of that object, as seen in Figure 6.

At this point, the position, rotation, and scale values of the object can be modified according to the researcher's preferences using the Inspector window.

To learn more about this process, use this [link](#).

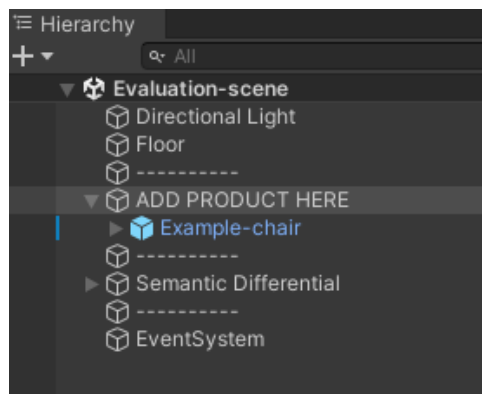
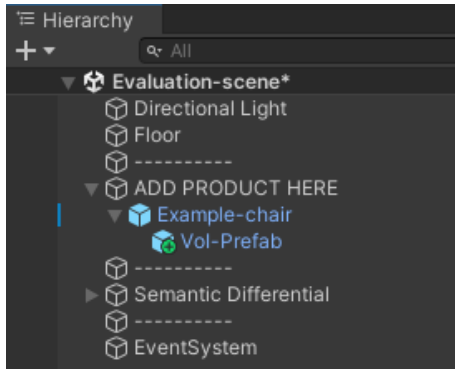


Figure 6. Adding a 3D Model to the Hierarchy in the "Evaluation-scene".

2.3. Configuring Volumes of Interest

In order to collect data on the product's visual focus, it is necessary to define specific Volumes of Interest (Vols). To do this, within the "03 Prefabs" folder, there is an object called "Vol-Prefab", which is pre-configured to function as a Vol. This object should be dragged into the hierarchy as a child of the product and duplicated (Ctrl + D) as many times as there are volumes of interest (for example, if I have a chair with four legs, a seat, and a backrest, I will have six Vol-Prefabs in the scene).

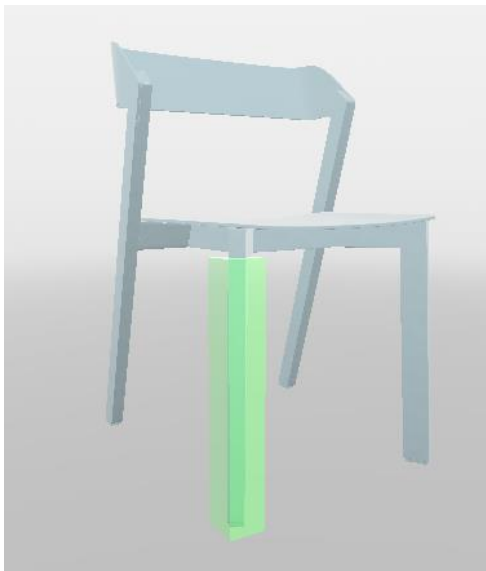
In Figure 7.a and Figure 7.b, you can see a Vol-Prefab recently added to the scene. If we select it, we can adjust its scale, position, and rotation values until it covers the part of the product we are interested in (Figure 7.c). Finally, in the hierarchy, it is necessary to update the name, as this will be the name that appears in the final data CSV. Additionally, the "Mesh Renderer" component should be turned off (Figure 7.d).



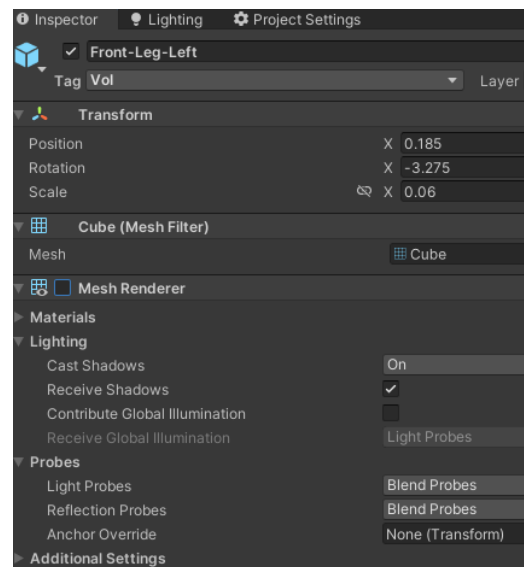
(a)



(b)



(c)



(d)

Figure 7. Vol in the scene.

2.4. Semantic Differential configuration

To gather the user's opinion about the displayed object, there is a panel that shows a series of semantic scales using a Likert-type scale from -3 to 3, with 0 being a neutral value. This component is called "Semantic Differential" in the hierarchy of the "Evaluation-scene".

Currently, there is only one example scale. However, this component can be duplicated (Ctrl + D) in the hierarchy and configured according to specific attributes. If we expand the "Semantic Differential – Canvas – Semantic scales" component, we will find the "SD#" component (Figure 8.a). If we want to gather opinions on six attributes (for example), we should duplicate the component six times and can rename them in a specific way to better identify them (Figure 8.b).

Additionally, if we expand each of these components, we will see one called "L-Adj" (which corresponds to the adjective on the left) and another called "R-Adj", corresponding to the adjective on the right. If we select it, we can change the adjective name in the TextMeshPro – Text (UI) component of the inspector (Figure 8.c).

Finally, we need to select the "Semantic Differential" component again in the hierarchy. In the inspector, search for the "Semantic Differential Responses (Script)" component and add each of the previously created semantic scales in the section labeled "Semantic Scales" (Figure 9). To add more items, click "+".

It is important that the semantic scales are added from top to bottom in the order they will appear. For example, SD1, which will be the first pair of adjectives shown to the user, should be placed in Element 0, and so on.

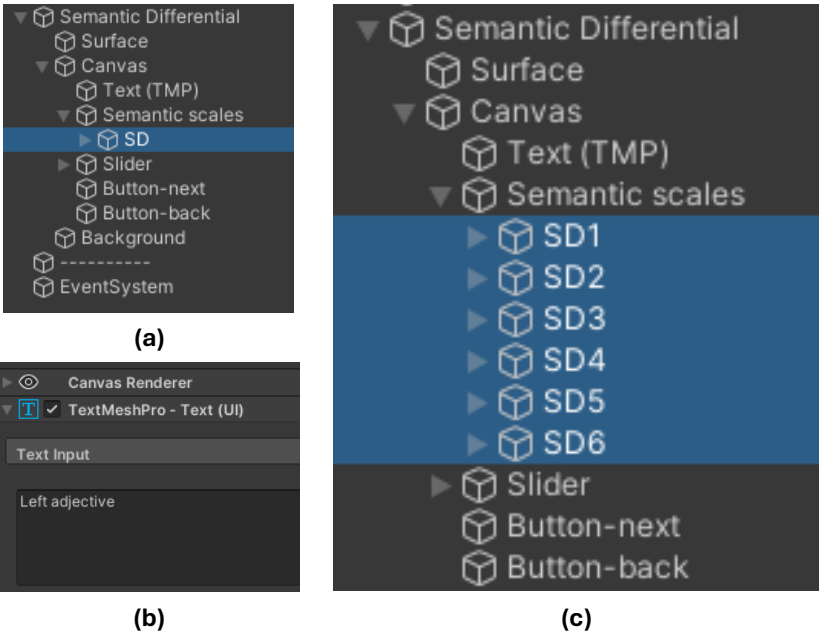


Figure 8. Semantic Differential configuration.



Figure 9. Adding semantic scales to the Semantic Differential Responses (Script).

◆ FUNCTIONING

1. Using the editor

If you are going to use the headset **connected to the computer**, this is your option.

First, you should follow the steps indicated here to use your HMD.

Before clicking Play, you will need to find the following objects in the Unity hierarchy to apply some changes:

- Manager (S-1): Uncheck the "To HMD" box in "Eye Tracking Recorder (Script)" and "Position Recorder (Script)" (within the Inspector) under "Export to HMD?" (Figure 10).
- Semantic Differential (S-2): Uncheck the "To HMD" box in "Semantic Differential Responses (Script)" (within the Inspector) under "Export to HMD?" (Figure 10).



Figure 10. Unmarked component to export the data to the streamingAssets folder of the Unity project.

2. Building an APK

If you want to use the headset in **standalone mode**, this is your option.

First, you need to build an APK (for more information, visit this [link](#)), but beforehand, you must check the option shown in Figure 10 in the same components mentioned in the previous section. This will ensure that the data is exported to the headset. It should look like Figure 11.

Once you have changed the previous configuration, go to Project settings – Meta. Click on Fix and Apply all in case there are any errors or recommended changes. Then, go to File – Build Settings, and in the Platform section, click on Android and then Switch Platform. This will allow you to build the APK.

When you click Build, you will need to give the application a name (the name by which you identify the APK on your PC, as it will appear as "VR-Based_Product-Evaluation-System" in the headset once installed).

Once you have the APK, install it on the headset. You will need to download the [Meta Quest Developer Hub](#) (MQDH) software and follow this [procedure](#). Once installed, you can use it.



Figure 11. Component marked to export the data to the headset.

◆ DATA ANALYSIS

If you have used the application through the editor, the CSV files have been saved in a folder called "streamingAssets" within the Unity project. If not, you will need to find the files in the installation folder of the application inside the headset (the path is similar to "QuestPro/Android/Data/AppName/").

Only in the latter case, copy all the CSV files and save them locally on your PC. You can save them in any folder, but make sure to remember the path, as you will need to use it later.

1. Installing Python

Navigate to the "05 Scripts – Data analysis" folder and run "python-3.13.2.exe". If you prefer a more up-to-date version, go to the official Python [website](#) to download it. In the installation window, check the options *Use admin privileges when installing py.exe* and *Add python.exe to PATH*.

2. Installing libraries

Open the Command Prompt (CMD). You can do this by typing "cmd" in the Windows search bar (Figure 12). Type each of the following lines one by one to install the necessary libraries:

- pip install matplotlib
- pip install pandas
- pip install seaborn

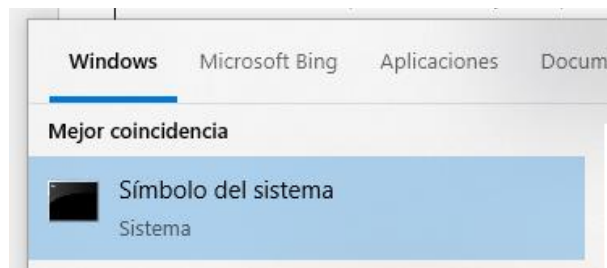


Figure 12. Command Prompt.

3. Scene snapshot

When drawing the subject's path and generating a heatmap of it on a top-down view of the scene, it is necessary to capture the scene. To do this, follow these steps:

1. Open the Unity project and go to S-2.
2. Add the "Snapshot" prefab to the hierarchy (Figure 13.a).
3. Select the prefab in the hierarchy and click the "CAPTURE" button within the inspector window (Figure 13.b).

This will generate a capture named "SceneSnapshot.png" within the project's streamingAssets folder.

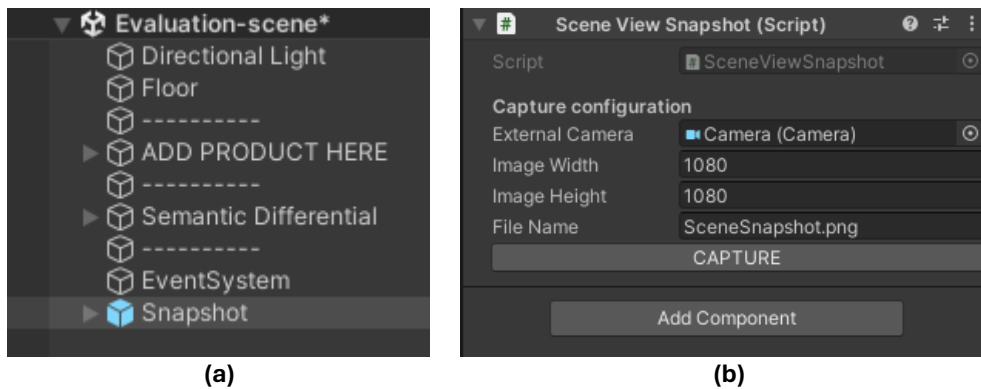


Figure 13. Procedure for capturing a top view of the S-2 scene.

4. Running interface.exe

In the "05 Scripts – Data analysis – dist" folder, there is an executable file named "interface.exe" that you need to run (Figure 13). First, you must specify the path where all the generated CSV files are stored by clicking on the "Select Folder" button.

Once specified, click on each of the buttons located at the bottom to manage the collected data (Position, Position heatmap, and Eye-tracking).

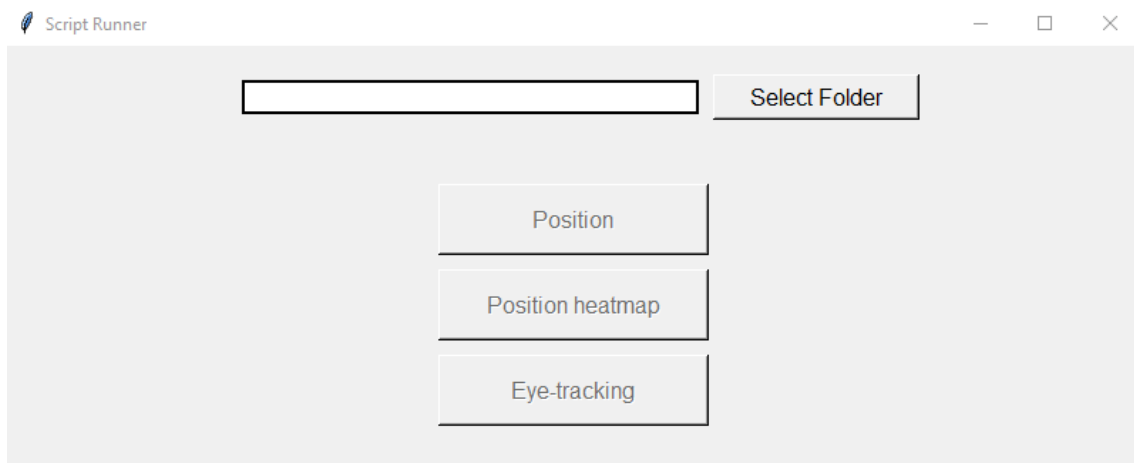


Figure 13. Executable interface for managing data obtained from the virtual environment.

5. Data review

All the images, as well as the final CSV file containing the Eye Tracking (ET) data ("final-data.csv"), are stored in the same folder where the original CSV files are located (the one you added in the previous section).